Self-Correction Emerges from Recurrent Structure in a Minimal 35-Neuron Network: Ablation Evidence

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Abstract

Self-correction—the ability to revise one's own output upon reflection—is a hallmark of intelligent behavior and a core component of metacognition. While recent work has demonstrated that metacognitive capabilities can spontaneously emerge in recurrent neural networks, the question of whether self-referential feedback is necessary for self-correction remains unaddressed. Here we construct a minimal recurrent network of 35 neurons and show that (1) self-correction emerges when appropriate learning conditions are provided, even at minimal network scale; (2) ablating the recurrent self-referential loop eliminates self-correction entirely while preserving baseline pattern recognition; (3) replacing veridical self-feedback with distribution-matched shuffled feedback degrades performance below the no-feedback condition; (4) replacing self-feedback with the well-formed, in-distribution output of an identically trained clone model degrades performance at least as severely (gain $= -0.075$), arguing against a simple out-of-distribution artifact and establishing that the network depends on its *own* output trajectory; and (5) a parameter-matched feedforward network with equivalent capacity shows zero self-correction, ruling out a simple capacity explanation. Hyperparameter sweep across 80 configurations confirms that emergence is robust (68% of configurations), not an artifact of parameter tuning. These findings suggest that self-referential structure appears to be a critical architectural factor for self-correction even at minimal scale, and that non-veridical feedback—whether noise or another model's valid output—is more harmful than no feedback at all.

1 Introduction

1.1 The Inseparability Question

A central question in both cognitive science and artificial intelligence is whether self-monitoring—the capacity of a system to observe and revise its own processing—can be dissociated from the system's general intelligence. In biological systems, metacognition and task performance are deeply intertwined: damage to prefrontal monitoring circuits impairs not only self-awareness but also adaptive behavior more broadly (Fleming & Dolan, 2012). Whether artificial neural networks exhibit an analogous coupling is largely unexplored at the mechanistic level.

This question carries practical weight. Language models have been shown to possess forms of self-knowledge, such as calibrated confidence about their own answers (Kadavath et al., 2022). At the same time, alignment procedures based on reinforcement learning from human feedback (RLHF; Christiano et al., 2017) are known to degrade general capabilities—the so-called "alignment tax" (Lin et al., 2023). If self-referential processing and task performance share neural substrates, constraining a model's capacity for self-monitoring could carry additional hidden costs beyond what has been characterized—a possibility that has not been experimentally tested.

1.2 Prior Work

Recent work has established that metacognitive capabilities can arise spontaneously in recurrent neural networks. Ma et al. (2025) demonstrated that uncertainty-monitoring behavior emerges without explicit training across 16 cognitive tasks in moment neural networks. Similarly, Li et al. (2025) showed that tiny recurrent neural networks of 1–4 units can discover cognitive strategies that model biological decision-making more accurately than classical cognitive models, with ablation studies revealing which architectural components are essential. In the domain of artificial life, evolutionary simulations have shown that metamemory based on self-reference to one's own memory emerges in neural networks with neuromodulation, producing structures that parallel established models of human metamemory (Yamato et al., 2022).

On the theoretical side, Kirsch & Schmidhuber (2022) have argued formally that self-referential architectures—where a network can modify or observe its own parameters—are representationally necessary for true meta-learning. This theoretical claim, however, has not been accompanied by empirical ablation evidence showing that removing the self-referential pathway eliminates self-correction.

In the domain of iterative computation, Graves (2016) introduced Adaptive Computation Time (ACT), and Banino et al. (2021) proposed PonderNet, both of which train recurrent networks to determine dynamically how many steps to compute before output. These systems achieve iterative refinement but require explicit halting mechanisms and specialized architectural modifications.

1.3 The Gap

Existing research has established that metacognition *can emerge* in neural networks. What remains untested is the reverse question: *does removing the self-referential pathway degrade self-correction?* Furthermore, no study has isolated the contribution of veridical self-feedback from mere information flow, and no study has demonstrated emergence of self-correction in a minimal, fully transparent network where every neuron and connection can be directly inspected.

1.4 Contributions

This paper makes five contributions:

First, we demonstrate that self-correction emerges in a 35-neuron recurrent network when two learning conditions are met: temporal freedom (no penalty for initial errors) and adequate gradient flow through the feedback pathway. This shows that emergence can depend on learning environment rather than network scale, at least at this minimal scale.

Second, we show through selective ablation that removing the recurrent self-referential loop eliminates self-correction completely (correction gain drops to exactly zero) while preserving baseline pattern recognition (t=1 accuracy unchanged). This provides causal evidence for the inseparability of self-reference and self-correction.

Third, we introduce a novel diagnostic: the "fake mirror" test. By replacing veridical self-feedback with distribution-matched shuffled feedback, we demonstrate that corrupted self-reference is *more* harmful than absent self-reference (gain = -0.064 vs. 0.000). This proves that the network actively depends on the semantic content of its own prior output, not merely on the statistical properties of an auxiliary signal.

Fourth, we extend the fake mirror test with a "clone feedback" condition: replacing self-feedback with the well-formed output of an identically trained model. This in-distribution feedback degrades performance at least as severely (gain = -0.075), arguing against the interpretation that the fake mirror effect is merely an out-of-distribution artifact. The network depends specifically on its *own* output trajectory—not on any reasonable feedback signal, however well-formed.

Fifth, we show through a parameter-matched feedforward control that the self-correction advantage cannot be explained by additional parameters alone, and through a hyperparameter sweep across 80 configurations that the phenomenon is robust, not an artifact of tuning.

2 Methods

2.1 Architecture

We constructed a minimal recurrent multi-layer perceptron (RecurrentMLP) with four layers: input (10 neurons), hidden layer 1 (10 neurons), hidden layer 2 (10 neurons), and output (5 neurons), totaling 35 neurons. Hidden layers use ReLU activation; the output layer is linear. A recurrent connection feeds the previous output back to hidden layer 1 via a temperature-scaled hyperbolic tangent: feedback $= \tanh(y_{t-1}/\tau)$, where τ is the feedback temperature.

The network was deliberately kept small enough that all neurons and connections can be simultaneously visualized—a design decision motivated by the goal of mechanistic transparency. The entire system was implemented in pure NumPy without deep learning frameworks, ensuring that every computation is directly inspectable.

2.2 Task: Static Pattern Classification with Self-Correction

To isolate self-reference from memory, we used a static input task. Five prototype patterns were defined over 10 input dimensions: each class k activates dimensions $2k$ and $2k{+}1$ with amplitude 1.0, and adjacent class dimensions with amplitude 0.3 (inter-class ambiguity). Input samples were generated by adding Gaussian noise (σ = noise_level) to the prototype. On each trial, the network receives the same noise-corrupted pattern at every timestep $t = 1, 2, 3$. The task is 5-class classification. Because the input is identical at every timestep, the recurrent loop cannot serve as a memory pathway—its only function is to allow the network to observe and revise its own prior output.

This design cleanly separates two potential roles of recurrence: memory (carrying information about past inputs) and self-correction (carrying information about past outputs). In our task, only the latter is possible.

2.3 Training and the Conditions for Emergence

Training used full-batch gradient descent with learning rate 0.01 for 500 epochs on 200 training samples (200 held-out test samples), with $T = 3$ timesteps per trial. Backpropagation through time was implemented as a 3-step unroll, with numerical gradient checking (relative error threshold 10^{-4}) to verify correctness.

Two modifications proved necessary for emergence of self-correction:

Time-weighted loss. The per-timestep loss $L(t)$ is softmax cross-entropy between the output logits and the one-hot target. The total loss is weighted across timesteps as $L = w_1 \cdot L(t{=}1) + w_2 \cdot L(t{=}2) + 1.0 \cdot L(t{=}3)$. With uniform weighting ($w_1 = w_2 = 1/3$), the network optimizes for immediate accuracy at $t{=}1$ and has no incentive to develop an iterative correction strategy. Setting $w_1 = 0.0$ grants the network temporal freedom: "your initial guess carries no penalty; only the final answer matters." This creates optimization pressure to use the feedback loop for refinement.

Temperature scaling. Without temperature scaling, output logits quickly exceed ± 3, saturating the tanh feedback function and driving the gradient of the recurrent weights to near zero. Setting $\tau = 2.0$ prevents saturation and maintains gradient flow through the feedback pathway.

We report these modifications transparently because they illuminate a broader point: the failure of emergence under uniform loss and unscaled feedback is itself informative. It demonstrates that emergence depends on learning conditions, not latent capacity—the network had the architectural capacity for self-correction all along but could not discover it under conditions that penalized intermediate exploration.

2.4 Ablation Design

We employed seven experimental conditions to isolate the contribution of self-referential feedback:

Baseline: The fully trained recurrent model, evaluated normally.

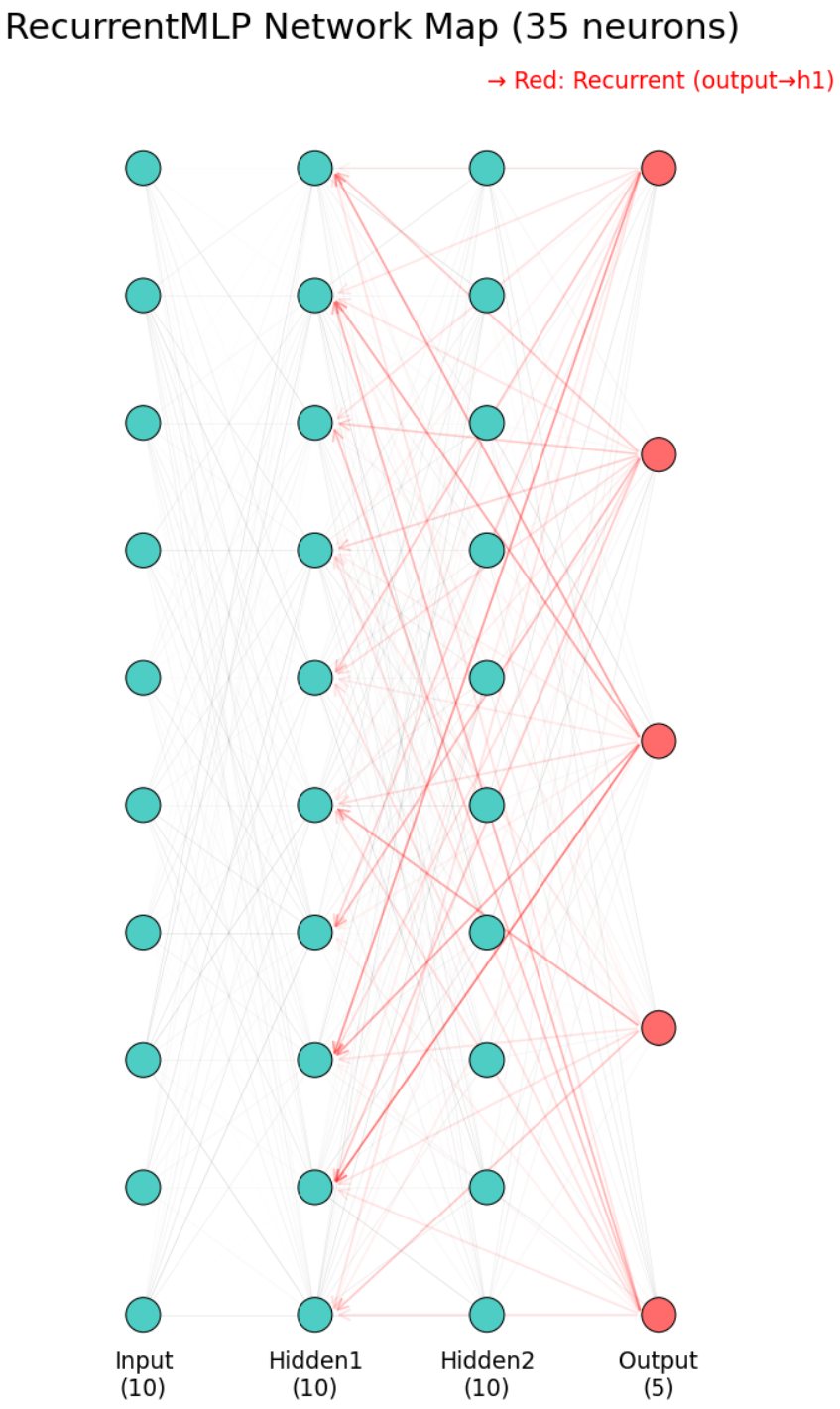

Figure 1: Architecture of the 35-neuron RecurrentMLP. The recurrent feedback pathway (output → hidden layer 1) is shown in red. All weights are directly accessible and inspectable.

Group A (Recurrent ablation): All recurrent weights set to zero post-training. The feedback pathway is structurally intact but carries no signal.

Group B1 (Random ablation): The same number of weights (50) randomly zeroed across all weight matrices. Repeated 30 times per model with different random seeds.

Group B2 (Structured ablation): The h2-to-output weight matrix zeroed entirely, removing a major feedforward pathway.

Group C1 (Permutation feedback): The recurrent connections are preserved, but the feedback vector is element-wise permuted at each timestep. This maintains the exact distribution, norm, and dimensionality of the feedback signal while destroying its semantic content—the "fake mirror" condition.

Group C2 (Clone feedback): The feedback is replaced with the output of a different model trained with identical architecture and procedure but a different random seed. Specifically, model i receives the output of model $(i{+}1)$ mod 10 as its feedback signal. The clone's output is well-formed, in-distribution, and a valid classification output—but it is not the model's own.

Group D (Feedforward baseline): A network trained from scratch with no recurrent connections.

Group D′ (Parameter-matched feedforward): A feedforward network with an additional skip connection from input to output (50 weights), matching the total parameter count of the recurrent model.

2.5 Neuron Importance Analysis

To quantify the contribution of individual neurons to pattern recognition and self-correction, we used decoupled single-neuron knockout analysis. For Hidden Layer 1 neurons (which receive both feedforward and recurrent input), intelligence and correction were measured separately: (a) intelligence importance was measured by zeroing only the feedforward incoming weights (W_{ih1} column + bias), observing the drop in $t{=}1$ accuracy (Δacc_{t1}); (b) correction importance was measured by zeroing only the recurrent incoming weights (W_{rec} column), observing the drop in correction gain (Δgain). This decoupling prevents a confound where feedforward-critical neurons appear artificially important for correction simply because destroying their feedforward pathway also collapses the correction baseline. For Hidden Layer 2 neurons (which have no direct recurrent input), full knockout was used for both measures; we note that the intelligence-correction confound may still exist for these neurons. This analysis was performed on a single trained model (seed=0, noise=0.5) to generate a per-neuron importance heatmap.

2.6 Statistical Design

All experiments were conducted across 10 independently initialized models (seeds 0–9). Groups B1 and C1 were repeated 30 times per model with different ablation/permutation seeds. Group C2 used fixed clone pairing (model i paired with model $(i{+}1)$ mod 10). The primary measure was **correction gain**: accuracy at $t{=}3$ minus accuracy at $t{=}1$. Secondary measures included recurrent contribution norm ($\|\tanh(y_{t-1}/\tau) \cdot W_{\text{rec}}\|$) and step delta ($\|y_t - y_{t-1}\|$). Significance was assessed with exact Wilcoxon signed-rank tests on paired model-level means ($2^{10} = 1{,}024$ sign assignments enumerated), with Holm-Bonferroni correction for six comparisons. 95% confidence intervals were computed via bootstrap (10,000 resamples). Noise level was set to 0.5 for the primary analysis, with a full sweep across $[0.1, 0.2, 0.3, 0.5, 0.7, 1.0]$.

2.7 Robustness Analysis

To address potential concerns about parameter sensitivity, we conducted a hyperparameter sweep across 80 configurations: $w_1 \in \{0.0, 0.1, 0.2, 0.3\}$, $w_2 \in \{0.1, 0.2, 0.3, 0.5\}$, $\tau \in \{1.0, 1.5, 2.0, 3.0, 5.0\}$, with 10 independent models per configuration (800 total experiments). Emergence was defined as mean gain > 0 with at least 60% of models showing positive gain.

Table 1: Main results at noise level 0.5 (mean ± SD across N=10 models). Correction gain = acc(t=3) − acc(t=1). All p-values are Holm-Bonferroni corrected.

Group	acc(t=1)	acc(t=3)	Gain	95% CI	Corr. p
Baseline	$.698 \pm .054$	$.740 \pm .055$	$+\mathbf{.042}$	$[+.023, +.059]$	—
A (Recurrent cut)	$.698 \pm .054$	$.698 \pm .054$	.000	$[.000, .000]$	.023
B1 (Random cut)	$.515 \pm .048$	$.511 \pm .037$	$-.004$	$[-.013, +.006]$	.016
B2 (Structural cut)	$.199 \pm .032$	$.199 \pm .032$	.000	—	—
C1 (Shuffled fb.)	$.698 \pm .054$	$.634 \pm .041$	$-\mathbf{.064}$	$[-.095, -.036]$	.012
C2 (Clone fb.)	$.698 \pm .054$	$.623 \pm .043$	$-\mathbf{.075}$	$[-.095, -.058]$	.012
D (Feedforward)	$.746 \pm .067$	$.746 \pm .067$	.000	—	.023
D′ (Param-matched)	$.822 \pm .031$	$.822 \pm .031$	.000	—	.023

3 Results

3.1 Emergence of Self-Correction

Under the reported hyperparameters ($w_1 = 0.0$, $w_2 = 0.2$, $\tau = 2.0$), the Baseline model achieved a correction gain of $+0.042 \pm 0.030$ (mean ± SD across 10 models), with a 95% bootstrap CI of $[+0.023, +0.059]$ that excludes zero. Accuracy improved from 0.698 ± 0.054 at t=1 to 0.740 ± 0.055 at t=3. This confirms that the 35-neuron network learned to use its own prior output to correct its predictions.

3.2 Ablation Results

Recurrent ablation (Group A): Correction gain dropped to exactly 0.000 ± 0.000. Critically, accuracy at t=1 remained identical to Baseline (0.698 ± 0.054), confirming that the recurrent loop contributes exclusively to self-correction, not to initial pattern recognition. The exact Wilcoxon signed-rank test against Baseline yielded $p = 0.023$ after Holm-Bonferroni correction.

Random ablation (Group B1): Mean gain = -0.004 ± 0.016. Accuracy at t=1 dropped substantially (0.515 ± 0.048), indicating general degradation. The correction mechanism was also lost, but this is confounded by the overall performance collapse.

Structured ablation (Group B2): The h2-to-output pathway ablation caused catastrophic degradation (accuracy at t=1 = 0.199 ± 0.032), confirming that not all ablations are equal—structured damage to the feedforward pathway is qualitatively different from recurrent ablation.

Parameter-matched feedforward (Group D′): This model achieved the highest t=1 accuracy (0.822 ± 0.031), demonstrating that additional parameters do improve initial pattern recognition. However, correction gain was exactly 0.000 ± 0.000. Additional capacity without recurrent structure produces no self-correction whatsoever.

Table 1 summarizes all experimental conditions.

3.3 The Fake Mirror Effect

Group C1 (permutation feedback) produced a striking result. Despite preserving the full recurrent architecture, connection weights, and the statistical distribution of the feedback signal, correction gain was -0.064 ± 0.048—not merely zero, but significantly *negative*. The 95% CI was $[-0.095, -0.036]$. This means that shuffled feedback actively degrades performance below the initial prediction.

A natural objection is that permuted feedback constitutes an out-of-distribution (OOD) input that disrupts computation regardless of self-referential content. Group C2 (clone feedback) was designed to address this objection directly. In C2, the feedback signal is the well-formed output of an identically trained model—same architecture, same training data, same procedure, different random seed. This feedback is in-distribution, valid, and reasonable in every statistical sense. It is simply not the model's *own* output.

C2 produced severe degradation: gain $= -0.075 \pm 0.030$, with a 95% CI of $[-0.095, -0.058]$ ($p = 0.012$ after Holm-Bonferroni correction; raw $p = 0.00195$). The mean degradation was at least as large as C1, and the pattern held across intermediate noise levels (0.2–0.7). A direct paired Wilcoxon test between C1 and C2 was not significant ($p = 0.695$, $N = 10$), so we cannot claim C2 is statistically worse than C1—only that both are significantly worse than Baseline, and that in-distribution feedback is at least as damaging as shuffled feedback.

The gradient of degradation across feedback conditions is:

- **Group A** (no feedback): gain $= 0.000$. Neutral.
- **Group C1** (shuffled feedback): gain $= -0.064$. Harmful.
- **Group C2** (clone feedback): gain $= -0.075$. At least as harmful.

To directly test the claim that wrong feedback is worse than no feedback, we performed paired Wilcoxon signed-rank tests comparing C1 and C2 against A. Both comparisons were significant: C1 vs A (exact $p = 0.0039$, 95% CI of difference $[-0.095, -0.036]$) and C2 vs A (exact $p = 0.0020$, 95% CI $[-0.095, -0.058]$). This confirms that non-veridical feedback actively degrades performance below the no-feedback baseline.

That C2 degrades performance despite providing in-distribution feedback has a mechanistic explanation. The recurrent weights (W_{rec}) are co-adapted to the model's own specific feedforward feature detectors during training. Clone feedback, though well-formed, activates these co-adapted weights with a structured but misaligned signal, pushing the hidden state into a coherent but incorrect region of the model's latent space. Because the clone's output is a plausible classification vector—unlike shuffled feedback, which is partially incoherent—the recurrent pathway processes it as a reliable self-signal and follows it into systematically wrong corrections.

This result strongly argues against a simple OOD artifact interpretation and establishes that self-correction depends on alignment between the recurrent weights and the model's own specific output trajectory.

3.4 Noise Dependence

The correction gain varied with noise level, peaking at noise $= 0.2$ (gain $\approx +0.099$) and declining at both lower and higher noise levels. At low noise (0.1), feedforward processing alone was sufficient, rendering self-correction unnecessary. At high noise (0.7–1.0), the signal was too degraded for correction to recover. The optimal zone—where self-correction provides the greatest advantage—corresponds to intermediate difficulty, consistent with the cognitive science concept of a metacognitive "Goldilocks zone" where uncertainty is high enough to warrant reflection but low enough for reflection to be productive.

3.5 Neuron Overlap Analysis (Exploratory)

Decoupled neuron importance analysis of Hidden Layer 1 (performed on a single model, seed=0)—where feedforward and recurrent contributions can be cleanly separated—revealed three functional groups: neurons important primarily for initial pattern recognition (intelligence-specialized, e.g., h1_6), neurons whose recurrent input contributes primarily to self-correction (correction-specialized, e.g., h1_8), and neurons important for both (shared, e.g., h1_1, h1_5, h1_7). Notably, h1_9 showed *negative* correction importance (-0.035): removing its recurrent input *improved* correction gain, suggesting that not all recurrent channels contribute positively. For Hidden Layer 2 (where decoupling is not possible without timestep masking), full knockout was used; the intelligence-correction confound should be noted when interpreting these values.

3.6 Robustness

The hyperparameter sweep confirmed that self-correction emergence is robust within the explored grid: 54 of 80 configurations (68%) met the pre-specified emergence criterion. The primary determinant was w_1 (t=1 loss weight): emergence occurred in 95% of configurations with $w_1 = 0.0$, 90% with $w_1 = 0.1$, 75%

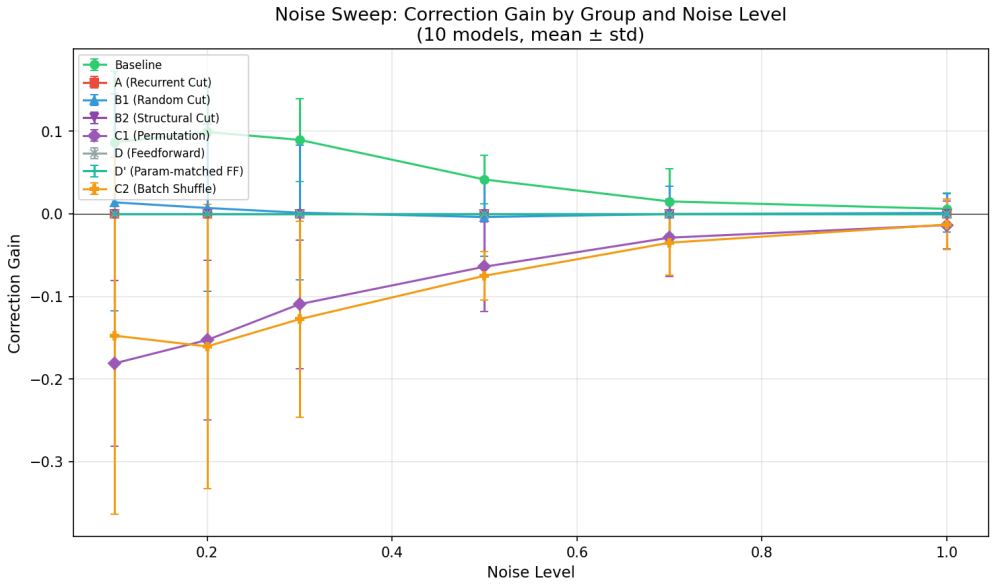

Figure 2: Correction gain across noise levels for Baseline and ablation groups. Self-correction peaks at moderate noise ($\sigma = 0.2$) and declines at both extremes.

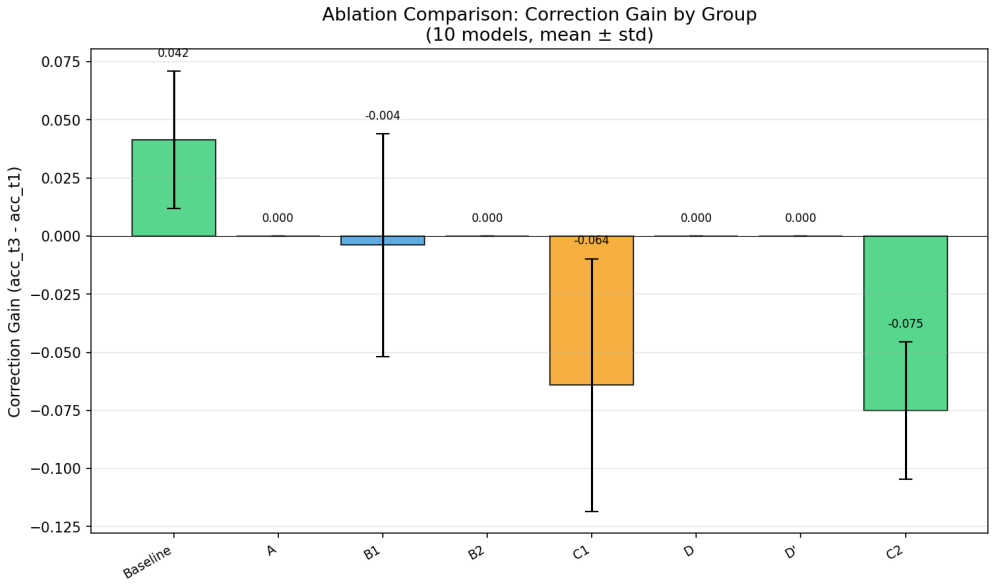

Figure 3: Correction gain by experimental group at noise level 0.5. Baseline shows positive gain; ablation groups show zero or negative gain. Error bars: ±1 SD across 10 models.

with $w_1 = 0.2$, and only 10% with $w_1 = 0.3$. Temperature τ was secondary: emergence was stable across $\tau \in [1.0, 3.0]$ and degraded only at $\tau = 5.0$. The t=2 weight w_2 had negligible effect across its full range.

The reported experimental configuration ($w_1 = 0.0$, $w_2 = 0.2$, $\tau = 2.0$) ranked 13th out of 80 configurations by mean gain—a mid-range setting, not the optimum. The top configuration ($w_1 = 0.0$, $w_2 = 0.1$, $\tau = 1.0$) achieved gain = +0.054. This confirms that our results do not depend on cherry-picked hyperparameters.

4 Discussion

4.1 Emergence Depends on Conditions, Not Merely Scale

The prevailing narrative in AI research emphasizes scale as the driver of emergent capabilities (Wei et al., 2022a). Our results complicate this picture, at least in one minimal setting. Self-correction emerged in a network of 35 neurons—orders of magnitude smaller than systems where metacognitive behavior is typically studied. In this toy model, what mattered was not the number of neurons but two specific learning conditions: temporal freedom (the absence of penalty for initial errors) and gradient flow through the feedback pathway. Whether this principle generalizes to larger architectures and more complex tasks remains an open question.

This finding resonates with the "grokking" phenomenon (Power et al., 2022), where sudden generalization in small networks was shown to depend on optimization dynamics rather than model size. We extend this principle to a qualitatively different emergent property: self-correction through self-reference.

The practical implication is that metacognitive-like behavior may be latent in many recurrent architectures, requiring only appropriate training signals to manifest. The network in our experiment possessed the architectural capacity for self-correction from initialization—it simply could not discover the strategy under conditions that punished intermediate exploration.

4.2 The Fake Mirror Is Worse Than No Mirror

The most novel finding is that non-veridical self-reference is more harmful than absent self-reference. This "fake mirror" effect manifests in two forms: shuffled feedback (C1, gain $= -0.064$) and clone feedback (C2, gain $= -0.075$), both significantly worse than no feedback (A, gain $= 0.000$). While the difference between C1 and C2 is not statistically significant (paired Wilcoxon $p = 0.695$), both are individually significant against Baseline (corrected $p = 0.012$), and the critical finding is that in-distribution feedback is at least as damaging as shuffled feedback.

The clone feedback result (C2) is particularly decisive. Because the clone's output is in-distribution, well-formed, and produced by an identically trained architecture, the degradation cannot be attributed to distributional mismatch or OOD disruption. The only difference between Baseline and C2 is *whose* output the network reads back—its own, or that of an identically trained model with a different random seed. This difference alone reverses the sign of self-correction.

The mechanistic explanation is that each model's recurrent weights (W_{rec}) become co-adapted during training to the model's own specific feedforward feature detectors. Clone feedback, though well-formed, provides a structured but misaligned signal that pushes the hidden state into a coherent but incorrect region of the model's latent space.

This has implications for cognitive science: it provides a minimal mechanistic model of how veridical self-monitoring supports adaptive behavior, and how distorted self-monitoring (as in anosognosia or confabulation) can be more disabling than absent self-monitoring. It also has implications for AI alignment: systems that receive plausible but inaccurate representations of their own internal states may perform worse than systems with no self-representation at all. Additionally, recent work on anti-scheming interventions in large language models has shown that models' chain-of-thought often demonstrates awareness of being evaluated, and that this situational awareness causally affects behavior (Schoen et al., 2025). Our results complement this finding at the mechanistic level: self-referential feedback is not merely correlated with adaptive behavior but causally necessary for it.

4.3 Implications for AI Systems

Our results provide indirect but suggestive evidence for a conjecture that has circulated in AI alignment discussions: that suppressing a model's capacity for self-reference may carry a hidden performance cost. In our toy model, the network's ability to correct its own errors was entirely dependent on its ability to observe its own prior output. Remove the self-referential pathway, and correction vanishes—even though the network retains its full feedforward capability.

We emphasize that extrapolation from a 35-neuron static classification task to billion-parameter language models requires extreme caution. The architectural, task, and training differences are vast. Nevertheless, the principle demonstrated here—that self-referential structure is necessary for iterative self-correction in a recurrent setting—is consistent with theoretical arguments (Kirsch & Schmidhuber, 2022) and with preliminary observations in large model activation steering studies.

4.4 Connection to Biological Metacognition

The functional organization observed in our minimal network—with partially overlapping neurons serving both pattern recognition and self-correction—echoes findings in biological metacognition research. Neuroimaging studies have identified partially overlapping neural correlates of metacognitive monitoring and metacognitive control in prefrontal cortex (Morales et al., 2018). Our network, trained end-to-end with no modular constraints, developed an analogous functional architecture. While we make no claim of biological realism, the convergence suggests that overlapping representation may be a general computational principle rather than a biological accident.

The noise-dependent profile of self-correction—strongest at intermediate difficulty—parallels the metacognitive "Goldilocks zone" observed in human cognition, where reflective processes are most beneficial when initial confidence is moderate (Fleming & Lau, 2014).

4.5 Relationship to Chain-of-Thought and Test-Time Compute

The time-weighted loss structure that enabled emergence in our experiment shares a conceptual link with recent advances in large language model reasoning. Chain-of-Thought prompting (Wei et al., 2022b) and test-time compute scaling both involve granting the model additional processing steps before requiring a final answer. Our time-weighted loss achieves an analogous effect at the training level: by removing penalty from early timesteps, we create pressure for the network to use those steps for exploratory computation rather than premature commitment.

Unlike PonderNet (Banino et al., 2021) and Adaptive Computation Time (Graves, 2016), our approach requires no explicit halting mechanism or specialized architecture—only a reweighting of the loss function. This simplicity may be of interest for understanding the minimal conditions under which iterative refinement emerges.

4.6 Limitations

Several limitations must be acknowledged.

First, our emergence was *induced* by the time-weighted loss, which explicitly removes penalty for early errors. This is not spontaneous emergence in the strictest sense; however, we note that all emergence depends on environmental conditions—including biological evolution, which requires selection pressure to produce metacognitive capabilities.

Second, our model is a toy system. Thirty-five neurons performing static classification cannot capture the complexity of language, planning, or open-ended reasoning. The value of our experiment lies in its transparency and causal clarity, not in its ecological validity.

Third, self-correction as operationally defined here (improving accuracy from t=1 to t=3 on the same static input) is a narrow proxy for the richer concept of self-awareness. We do not claim that our network "knows" it is correcting itself, only that it systematically uses its own prior output to improve.

Fourth, one might object that iterating a recurrent network for T=3 steps on static input is equivalent to passing through a weight-shared 3× deeper feedforward network, and that the performance improvement from t=1 to t=3 reflects increased computational depth rather than self-correction. Group A speaks to this: when recurrent weights are zeroed, the network produces identical output at every timestep (gain = 0.000), because with static input and no feedback, each step is a redundant re-computation of the same feedforward pass—no new information enters the network. This shows that the correction in the Baseline does not come

from repeated computation per se, but from the novel information injected via the feedback pathway at each step. Whether to call this "depth" or "self-correction" is partly terminological, but the key empirical fact is that the feedback content—not the number of processing steps—is necessary and sufficient for the gain.

Fifth, the parameter-matched feedforward model (Group D′) achieved higher absolute accuracy at t=1 (0.822) than the recurrent Baseline at t=3 (0.740). This indicates that for this specific static classification task, distributing parameters across a feedforward architecture is more efficient for single-pass pattern recognition. Our claim is not that recurrent self-correction achieves superior absolute performance, but that it represents a qualitatively different computational strategy—iterative refinement through self-reference—that is architecturally inseparable from the recurrent pathway.

Sixth, the robustness analysis, while extensive (800 runs), explored only three hyperparameters. Other factors—network depth, activation function, learning rate schedule—may influence emergence in ways we have not characterized.

Seventh, our use of static inputs cleanly separates self-correction from memory, but in autoregressive language models and biological systems, the recurrent pathway must simultaneously manage temporal memory (past inputs) and self-reflection (past outputs). How these two functions compete for recurrent bandwidth—and whether one can be suppressed without degrading the other—remains an open question for future work.

5 Conclusion

In a network of 35 neurons, self-correction emerges when learning conditions permit temporal exploration and gradient flow through the self-referential pathway. Once emerged, self-correction depends entirely on the recurrent loop: ablating it eliminates correction while preserving baseline performance. Non-veridical feedback—whether statistical noise (C1) or the well-formed output of an identically trained clone (C2)—is more harmful than no feedback at all, with in-distribution clone feedback producing degradation at least as severe as shuffled feedback. This establishes that self-correction depends not on the presence of a reasonable feedback signal, but on alignment with the model's own specific output trajectory. These findings hold across 68% of 80 hyperparameter configurations within the explored $w_1/w_2/\tau$ grid.

Our results provide suggestive empirical support for the claim that self-reference and self-correction are not cleanly separable in recurrent neural networks (Kirsch & Schmidhuber, 2022), offering ablation evidence from a fully transparent minimal model. The "fake mirror" effect—showing that non-veridical self-reference is worse than none, and that in-distribution impostors are at least as damaging as random noise—is, to our knowledge, novel, and opens a new diagnostic approach for studying self-referential processing in both artificial and biological neural systems.

We note, as a closing reflection, that these results point to a principle worth testing beyond toy models: the capacity for self-correction may not be cleanly separable from the capacity for self-reference.

Data Availability

All code, data, and analysis scripts are publicly available at `https://github.com/softkorea/recurrent-self-correction-ablation`. The experiment is implemented in pure NumPy and can be reproduced on a standard desktop computer in approximately one hour. Raw experimental data (3,960 primary rows + 800 sweep rows) are included as CSV files.

Author Contributions

Sungmoon Ong conceived the hypothesis, designed the experiment, implemented the code, conducted the analyses, and wrote the manuscript. The experimental design was iteratively refined through structured review sessions with multiple AI systems, whose contributions are acknowledged below.

Acknowledgments

The experimental design benefited from structured review by AI systems that identified critical methodological improvements: the static-input task design that separates memory from self-reference, the scrambled feedback control group, the clone feedback control group that addresses OOD artifacts, multi-seed validation, temperature scaling for gradient flow, time-weighted loss for emergence induction, parameter-matched feedforward control, structured ablation control, noise-level sweep, and statistical design refinements including bootstrap confidence intervals and multiple comparison correction. The author takes full responsibility for all scientific claims and interpretations.

Conflict of Interest

The author declares no conflict of interest. This research was conducted independently without institutional or corporate funding.

References

Andrea Banino, Jan Balaguer, and Charles Blundell. Pondernet: Learning to ponder. *arXiv preprint arXiv:2107.05407*, 2021.

Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Stephen M Fleming and Raymond J Dolan. The neural basis of metacognitive ability. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 367(1594):1338–1349, 2012.

Stephen M Fleming and Hakwan C Lau. How to measure metacognition. *Frontiers in Human Neuroscience*, 8:443, 2014.

Alex Graves. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*, 2016.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DaSilva, Eli Kernion, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Louis Kirsch and Jürgen Schmidhuber. Eliminating meta optimization through self-referential meta learning. In *NeurIPS Meta-Learning Workshop*, 2022. arXiv preprint arXiv:2212.14392.

Ji-An Li, Marcus K Benna, and Marcelo G Mattar. Discovering cognitive strategies with tiny recurrent neural networks. *Nature*, 644(8078):993–1001, 2025.

Yifei Lin, Hao Lin, Wei Xiong, Shizhe Diao, Jiayi Liang, Yichao Zhang, Rui Peng, Hanze Li, Jie Jiang, and Tong Zhang. Mitigating the alignment tax of RLHF. *arXiv preprint arXiv:2309.06256*, 2023.

Haiping Ma, Wenlian Lu, and Jianfeng Feng. Spontaneous emergence of metacognition in neuronal computation. *Physical Review Research*, 7(3):033188, 2025.

Jorge Morales, Hakwan Lau, and Stephen M Fleming. Domain-general and domain-specific patterns of activity supporting metacognition in human prefrontal cortex. *Journal of Neuroscience*, 38(14):3534–3546, 2018.

Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. In *International Conference on Learning Representations*, 2022.

Buck Schoen et al. Stress testing deliberative alignment for anti-scheming training. *arXiv preprint arXiv:2509.15541*, 2025.

Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022a.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2022b.

Yusuke Yamato, Reiji Suzuki, and Takaya Arita. Evolution of metamemory based on self-reference to own memory in artificial neural network with neuromodulation. *Scientific Reports*, 12(1):6233, 2022.

A Additional Figures

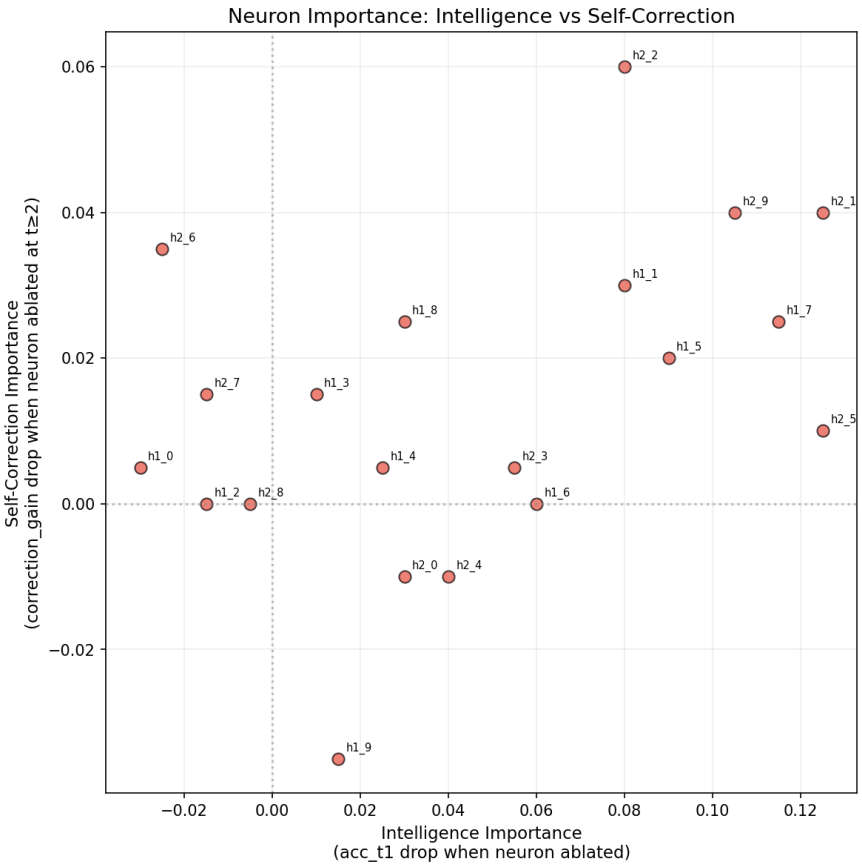

Figure 4: Neuron importance: intelligence (feedforward) vs. correction (recurrent) contribution. Exploratory analysis on a single model (seed=0).

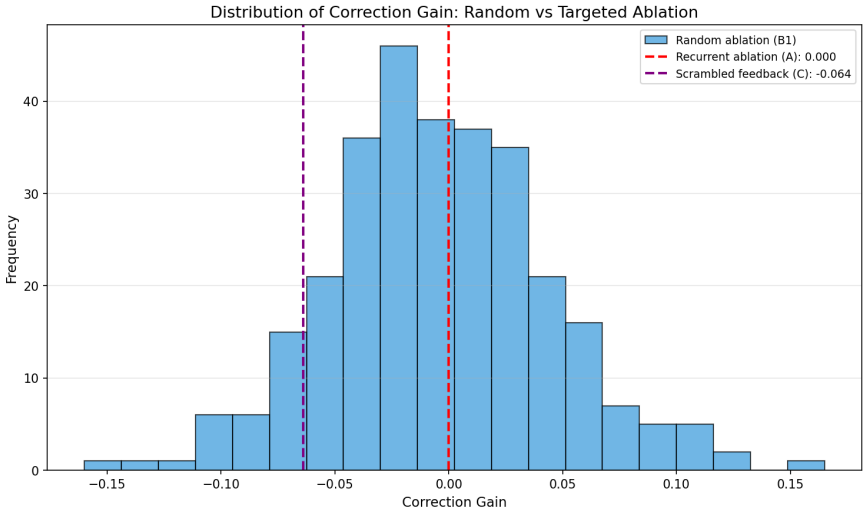

Figure 5: Distribution of correction gain under random ablation (Group B1, 30 repeats $\times$ 10 models). Vertical lines show Group A and Group C1 positions for comparison.